

51. What is a list in Python?

Answer:

A list is a collection of items stored in a single variable.

Lists are **ordered, changeable (mutable), and allow duplicate values.**

Example:

```
fruits = ["apple", "banana", "mango"]
```

```
print(fruits)
```

Output:

```
['apple', 'banana', 'mango']
```

Features of lists:

- Ordered
- Mutable (can be changed)
- Allows duplicate values
- Can store different data types

52. How do you add elements to a list?

Answer:

We can add elements using:

1. append()

Adds an item at the end of the list.

```
numbers = [1, 2, 3]
```

```
numbers.append(4)
```

```
print(numbers)
```

Output:

```
[1, 2, 3, 4]
```

2. insert()

Adds an item at a specific position.

```
numbers.insert(1, 10)
```

```
print(numbers)
```

Output:

```
[1, 10, 2, 3, 4]
```

53. What is the difference between append(), extend(), and insert()?

Answer:

Method	Purpose	Example
append()	Adds one item at the end	list.append(5)
extend()	Adds multiple items	list.extend([6,7])
insert()	Adds item at a specific index	list.insert(1,5)

Example:

```
a = [1, 2]
```

```
a.append(3)
```

```
a.extend([4, 5])
```

```
a.insert(0, 10)
```

```
print(a)
```

Output:

```
[10, 1, 2, 3, 4, 5]
```

54. How do you remove elements from a list?

Answer:

List elements can be removed using:

remove()

Removes a specific value.

```
a = [1, 2, 3]
```

```
a.remove(2)
```

```
print(a)
```

Output:

```
[1, 3]
```

pop()

Removes an element using index.

```
a.pop(0)
```

clear()

Removes all elements.

```
a.clear()
```

55. What is list slicing in Python?

Answer:

List slicing is used to get a specific part of a list.

Syntax:

```
list[start:end]
```

Example:

```
numbers = [10, 20, 30, 40, 50]
```

```
print(numbers[1:4])
```

Output:

```
[20, 30, 40]
```

Explanation:

- Start index is included.

- End index is excluded.